

Methodology (Corrected Mathematical Models and Algorithms)

Hand Detection Algorithm

The hand detection algorithm employs skin color segmentation using a transformation from the RGB color space to the YUV color space. The chrominance components are computed as:

$$\begin{aligned} Y &= 0.299R + 0.587G + 0.114B \\ U &= -0.147R - 0.289G + 0.436B \\ V &= 0.615R - 0.515G - 0.100B \end{aligned}$$

where R, G, and B represent the red, green, and blue color components, respectively. The U and V channels are used for skin tone segmentation as they are less sensitive to illumination changes.

Feature Extraction Using Histogram of Oriented Gradients (HOG)

Gradient components in the horizontal and vertical directions are computed as:

$$\begin{aligned} G_x(i, j) &= I(i+1, j) - I(i-1, j) \\ G_y(i, j) &= I(i, j+1) - I(i, j-1) \end{aligned}$$

The gradient magnitude and orientation are then calculated as:

$$\begin{aligned} M(i, j) &= \sqrt{G_x(i, j)^2 + G_y(i, j)^2} \\ \theta(i, j) &= \arctan(G_y(i, j) / G_x(i, j)) \end{aligned}$$

where $M(i, j)$ represents the gradient magnitude and $\theta(i, j)$ represents the gradient orientation at pixel (i, j) .

Convolutional Neural Network (CNN) Model

The output of a convolutional layer is defined as:

$$Y_{l+1}(i, j) = f\left(\sum_m \sum_n \sum_k W_l(m, n, k) \cdot X_l(i+m, j+n, k) + b_l\right)$$

where X_l is the input feature map of layer l , W_l represents the convolutional kernel weights, b_l is the bias term, $f(\cdot)$ is the activation function (ReLU), and the summations are taken over the spatial dimensions and input channels.

Dynamic Time Warping (DTW)

Dynamic Time Warping is used for temporal gesture sequence matching. The accumulated distance matrix is computed as:

$$D(i, j) = d(p_i, q_j) + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\}$$

where $d(p_i, q_j)$ is the Euclidean distance between elements p_i and q_j of two gesture sequences.

Classification Performance Metrics

System performance is evaluated using standard classification metrics:

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$\text{F1-Score} = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$$

where TP represents true positives, FP represents false positives, and FN represents false negatives.